Week 5 lecture 1

Programatic Polynomials

What is a Polynomial

- Poly: from the greek word for "many", POLÚS.
- nomial: from the greek NOMÓS, meaning "portions" or "parts"

What is a Polynomial

Denoting an equation with many parts

$$\underbrace{x^2} + \underbrace{x} + \underbrace{2}$$

What is a Polynomial

- An equation with multiple parts
- Coefficients accompanying each part

What is a Coefficient

Scaler multiple of any variable term in an equation

$$3x^2 + 2x + 1$$

3 2 1

$$25x^3 + 4x^2 + 9x^1 + 10x^0$$

25 4 9 10

“When Euler died, he simply said I am finished and collapsed, to which someone in the audience muttered darkly" Another conjecture of Euler is proved”

–Paul Erdős

Encoding polynomials

- Common polynomials have special structures
- Using that structure we can encode them onto a computer

Structure

- Concrete structures
- Structures $\Longrightarrow$ sequential problem solving
  - Think of the GJE algorithm

Structure of Polynomials

- Sequence $\{\ldots + x^2 + x^1 + x^0\}$
- Adding coefficients $\{\ldots + c_2x^2 + c_1x^1 + c_0x^0\}$
- Encoding $\{\ldots, c_2, c_1, c_0\}$

Coefficient row vector

- PN: $5x^2 + 4x^1 + 10x^0$
- CRV: [5,4,10]
  - Degree left associative

Synonyms

- Coefficient row vector
- Coefficient vector
- Coefficient array

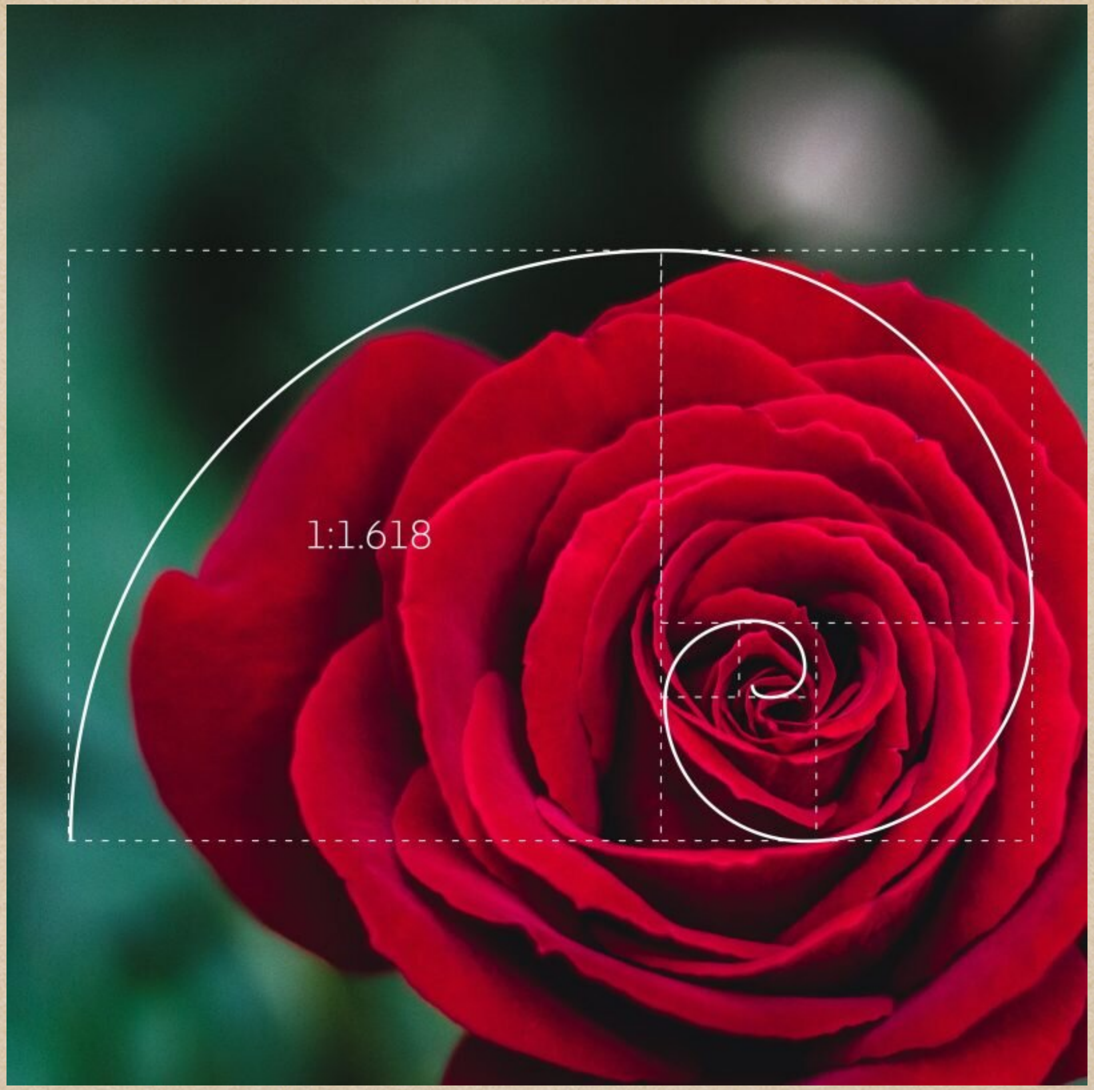
MATLAB

- *roots*(< *coef* >)
- *plot*(< *coef* >)

$\vdots$

"And it became clear to me that it's to build something smarter than myself, which will build something even smarter, et cetera, et cetera, and eventually colonize and transform the universe, and make it intelligent."

–Jürgen Schmidhuber



Statistics

Statistical analysis

- Interpolation
- Regression

Interpolation

- Fitting fxns to data to fill in internal data
- Using: $interp1(< data >)$

Regression

- Fitting fxns to data to fill in external data
- Using: *polyfit*(< *data* >)

Why?

- Discrete space
- Real world starts with data
- Cost of estimation << Cost of experimentation

Questions?